

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“JNANA SANGAMA”, BELAGAVI -590 018.



III Semester

PROJECT WORK
[MPRJ304]

Report on

MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System

Submitted in partial fulfilment of the requirements for the award of the degree of MCA

Department of Master of Computer Applications

Submitted By

Devaraju K G

1RN24MC021

Under the Guidance of

Dr. Nagesh B S

Associate Professor and Head, Department of MCA



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Channasandra, Dr. Vishnuvardhan Road, Bengaluru - 560 098

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Department of Master of Computer Applications



CERTIFICATE

This is to Certify that **Mr. Devaraju K G** bearing USN:1RN24MC021, student of **3rd SEM MCA** has satisfactorily completed the **Project work (MPRJ304)** entitled “**MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System**” in the academic year **2025-2026** as prescribed by VTU for 3rd Semester of Master of Computer Applications.

Internal Guide

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DECLARATION

I, **Devaraju K G** bearing **USN:1RN24MC021** student of **3rd SEM**, Department of MCA, RNS Institute of Technology, Bengaluru-98, declare that the **Project work** entitled “**MATA RAKSHA – AI Enhanced Biometric Blockchain Voting system**” is a record of the original work done by me under the guidance and supervision of **Dr. Nagesh B S**, Associate Professor and Head, Department of MCA, RNSIT and this project work has not formed the basis of any Degree/Diploma/Fellowship or similar title to any candidate of any university.

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Signature of the candidate

ACKNOWLEDGEMENT

I profoundly thank the **Management of RNS Institute of Technology** for their commitment to ensuring students have access to the best facilities and academic opportunities. Their constant support has made a major difference in the execution of this project.

I would like to express my sincere gratitude to **Dr. M K Venkatesha, The Director RNSIT**, for his constant support, encouragement, and guidance throughout the duration of this project.

I am extremely grateful to **Dr. Ramesh Babu H S, Principal RNSIT**, for providing the necessary resources and fostering an environment conducive to academic excellence. His leadership and vision have been a constant source of inspiration.

I would like to express my sincere appreciation to **Dr. Nagesh B S, Associate professor and HoD, Department of MCA**, for his guidance and for ensuring that I had the proper direction in my academic endeavours. His encouragement throughout this project has been invaluable.

I would like to sincerely thank my guide, **Dr. Nagesh B S, Associate professor and HoD**, for his invaluable guidance, continuous support, and expert advice throughout this project. His encouragement and constructive feedback have played a significant role in the successful completion of this work.

I express my truthful thanks to **Ms. Roopa H.M, Assistant Professor, Coordinator**, Department of MCA for her valuable support.

Finally, I take this opportunity to extend our earnest gratitude and respect to our parents, teaching & technical staff of the department, the library staff and all our friends, who have directly or indirectly supported me during the period of my project work.

Devaraju K G

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ABSTRACT

Traditional voting systems face challenges such as impersonation, lack of transparency, and delayed result processing. This project, **MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System**, proposes a secure and transparent digital voting solution by integrating biometric fingerprint authentication with blockchain technology.

The system verifies voters using fingerprint-based biometric matching, ensuring that only eligible voters can cast a vote. Each vote is securely recorded on a blockchain, providing immutability, transparency, and resistance to tampering. To maintain data privacy, biometric information is stored as encrypted hashes rather than raw biometric data.

The blockchain layer is implemented using smart contracts and demonstrated through a private blockchain environment to record vote transactions and enable real-time result counting. A role-based architecture is used to manage administrators, election officers, and voters effectively.

The proposed system enhances election integrity, reduces fraudulent practices, and demonstrates a scalable and cost-effective approach for secure digital voting in institutional and controlled election environments.

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Chapter-1

INTRODUCTION

1.1. Aim/objectives of Project work

The aim of the project is to design, develop, and implement a secure, transparent, and reliable digital voting system by combining biometric fingerprint authentication with blockchain technology. This phase focuses on converting the conceptual design into a fully working prototype and validating its functionality through real-time demonstrations.

The detailed objectives are as follows:

- To develop a biometric-based voter authentication mechanism that ensures only authorized voters are allowed to vote.
- To prevent impersonation, duplicate voting, and unauthorized access during the election process.
- To securely handle biometric information by storing fingerprint data in hashed form, ensuring voter privacy and compliance with data security principles.
- To implement a blockchain-based vote recording system where each vote is stored as an immutable and verifiable transaction.
- To design and implement smart contracts that control vote casting, transaction validation, and result computation.
- To create role-based dashboards for different election authorities such as Administrator, Registrar, and Election Officer.
- To enable real-time vote counting and automatic result declaration using blockchain records.
- To evaluate the feasibility of a cost-effective and scalable digital voting model suitable for institutional elections and pilot implementations.

1.2 Project Description

MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System is a secure digital voting platform developed to address the limitations of traditional and electronic voting systems. Conventional voting methods often face challenges such as voter impersonation, booth capturing,

manual errors in vote counting, and lack of transparency. This project aims to overcome these challenges by leveraging modern technologies.

The system begins with voter registration, where the registrar captures the voter's fingerprint using a biometric scanner. Instead of storing the raw fingerprint image, the system processes the fingerprint and generates a unique cryptographic hash, which is stored securely in the database. This approach ensures that sensitive biometric data is protected and cannot be misused.

During the voting phase, the election officer verifies the voter by matching the live fingerprint input with the stored hash. Once verification is successful, the voter is granted access to the voting interface. The voter selects a candidate, and the vote is recorded on the blockchain as a transaction executed through a smart contract.

Each vote becomes a permanent and tamper-proof record within the blockchain. Since blockchain data is immutable, it eliminates the possibility of vote manipulation or deletion. The blockchain also enables transparent and verifiable vote counting, as results are computed directly from the recorded transactions.

The system follows a role-based architecture:

- **Administrator:** Creates elections, manages candidates, and monitors blockchain transactions.
- **Registrar:** Registers voters and enrolls biometric data.
- **Election Officer:** Verifies voters and oversees the voting process.

A private blockchain environment is used to demonstrate blockchain concepts such as block creation, transaction validation, and smart contract execution. This approach allows real-time demonstration without the high costs associated with public blockchain networks.

1.3 Scope of the Project

The scope of the MATA RAKSHA project defines the functional and operational boundaries of the system in its current implementation.

In-Scope Features

The project scope includes:

-
- Biometric-based voter registration and verification using fingerprint authentication.
 - Secure handling of biometric data using hashing techniques.
 - Blockchain-based vote recording with immutability and transparency.
 - Smart contract-controlled vote casting and counting mechanisms.
 - Role-based access control for election authorities.
 - Real-time vote counting and result declaration.
 - Prevention of duplicate voting and unauthorized access.
 - Secure local data storage for voter and election information.

Operational Scope

- The system is designed for institutional elections, pilot studies, and controlled environments.
- The application operates as a desktop-based system.
- Blockchain operations are demonstrated using a private blockchain simulator, ensuring cost-effectiveness while preserving blockchain functionality.

Out-of-Scope (Current Phase)

The following features are not included in the current phase:

- Deployment on public blockchain networks.
- Large-scale national or state-level election implementation.
- Online remote voting over the internet.
- Mobile or web-based voting platforms.

Future Scope

Future enhancements may include:

- Integration with public blockchain networks for large-scale deployment.
- Advanced privacy mechanisms such as Zero-Knowledge Proofs (ZKP).
- Multi-factor biometric authentication.
- Web and mobile application support.
- Integration with government identity systems under regulatory approval.

Chapter-2

LITERATURE SURVEY

2.1 Existing System and Proposed System

Existing Systems:

The existing voting systems used today mainly include paper-based voting, electronic voting machines (EVMs), and limited forms of digital or online voting platforms. While these systems have improved over time, they still face several critical challenges related to security, transparency, and trust.

In traditional paper-based voting systems, voter identity verification is mostly manual and dependent on physical documents. This process is time-consuming and vulnerable to impersonation, duplicate voting, and human errors. Vote counting in such systems is slow and prone to manipulation during transportation and counting stages.

Electronic Voting Machines improve efficiency but still rely heavily on centralized control. Voter authentication is generally done using identity cards, which can be misused or forged. Since the vote data is stored in centralized systems, it becomes a potential target for tampering, unauthorized access, or internal manipulation. Additionally, voters and observers cannot independently verify the integrity of the recorded votes, which reduces transparency.

Existing online voting systems, where implemented, often depend on centralized servers and basic authentication mechanisms such as passwords or OTPs. These systems are vulnerable to cyberattacks, data breaches, and privacy concerns. Moreover, most existing systems store sensitive voter information, including biometric data, in plain or reversible formats, increasing the risk of data misuse.

Overall, the existing systems lack strong voter authentication, end-to-end transparency, tamper resistance, and secure handling of sensitive data, which are essential for building trust in digital elections.

Proposed System:

The proposed system, MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System, introduces a secure, transparent, and decentralized approach to digital voting by integrating biometric fingerprint authentication with blockchain technology.

In the proposed system, voter authentication is performed using fingerprint-based biometric verification, ensuring that only legitimate and registered voters can participate in the election. To protect voter privacy, the system does not store raw fingerprint images. Instead, fingerprint data is converted into a secure cryptographic hash, making it impossible to reconstruct the original biometric information.

Once a voter is successfully authenticated, the vote is recorded as a blockchain transaction using smart contracts. Each vote becomes an immutable and tamper-proof record stored within a blockchain block. This ensures that votes cannot be altered, deleted, or duplicated after being cast.

The proposed system also introduces a role-based architecture, where different users have clearly defined responsibilities:

- **Administrator** manages elections, candidates, and blockchain setup.
- **Registrar** handles voter registration and biometric enrollment.
- **Election Officer** verifies voters and oversees the voting process.

Vote counting and result declaration are performed automatically by analyzing blockchain transactions, enabling real-time and error-free result generation. The use of blockchain enhances transparency, as every vote is verifiable while maintaining voter anonymity.

The system is implemented using a private blockchain environment to demonstrate real-world blockchain behavior in a cost-effective manner. This approach maintains the same architectural principles as public blockchains while making the system suitable for academic and institutional use.

Key Advantages of the Proposed System

- Strong biometric-based voter authentication
- Tamper-proof vote storage using blockchain
- Enhanced transparency and trust
- Secure handling of sensitive voter data

- Automated and accurate vote counting

2.2 Tools and Technology Used

The MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System is developed using a combination of frontend technologies, backend programming tools, biometric hardware, blockchain frameworks, and database systems. These technologies collectively ensure secure voter authentication, transparent vote recording, and reliable result processing.

Frontend (User Interface Layer)

- **CustomTkinter (Python GUI Framework)**

CustomTkinter is used to design the desktop-based graphical user interface of the system. It provides a modern and responsive interface for different user roles such as Administrator, Registrar, Election Officer.

The frontend handles:

- User login and role-based dashboard display
- Election creation and management screens
- Voter registration and verification interfaces
- Voting interface for casting votes
- Display of election status and results

CustomTkinter ensures a clean, user-friendly, and interactive interface suitable for institutional usage.

Backend (Application Logic)

- **Programming Language:**

Python – used to implement core system logic including biometric verification, election workflow, blockchain interaction, and database management.

- **Development Tool:**

Python IDLE – used for writing, executing, and debugging Python code during development.

Biometric Technology

- **Fingerprint Scanner:**

SecuGen Hamster Pro 20 – used to capture fingerprint data during voter registration and verification.

- **Biometric SDK:**

SecuGen SDK (based on .NET Framework) – used to interface with the fingerprint device and acquire biometric data securely.

- **Biometric Security:**

Captured fingerprint data is converted into **hashed values**, ensuring that raw biometric images are never stored in the system.

Blockchain Technology

- **Smart Contract Platform:**

Ethereum Smart Contracts (Solidity) – used to define election rules, candidate registration, vote casting, and result computation.

- **Blockchain Environment:**

Ganache (Private Blockchain Simulator) – used to deploy and test smart contracts, simulate block creation, and record vote transactions in a cost-effective environment.

- **Blockchain Integration:**

Web3.py – used to enable communication between the Python backend and blockchain smart contracts.

Database Technology

- **Database:**

SQLite – used for secure local storage of voter details, hashed biometric data, election records, and system logs.

Security and Access Control

- **Cryptographic Hashing** – used to protect biometric and sensitive data.
- **Role-Based Access Control (RBAC)** – ensures only authorized users can access specific system functions.

Monitoring and Testing Tools

- **Web Browser (Chrome / Firefox)** – used for monitoring blockchain activity through the Ganache interface during testing and demonstration.

2.3 Hardware and Software requirements

Hardware Requirements:

Table 2.1 Hardware Requirements

Component	Specification
Processor	Intel i5 or equivalent
RAM	Minimum 8 GB
Storage	At least 256 GB HDD/SSD
Biometric Device	SecuGen Hamster Pro 20 Fingerprint Scanner
Network	Local system / Internet (for blockchain simulation)

Software Requirements:**Table 2.2 Software Requirements**

Component	Specification
Operating System	Windows 10 / Linux
Programming Language	Python 3.9 or higher
Frontend Framework	CustomTkinter
Development Tool	Python IDLE
Blockchain Environment	Ganache
Smart Contract Language	Solidity
Blockchain Library	Web3.py
Database	SQLite
Biometric SDK	SecuGen SDK (.NET Framework)
Web Browser	Chrome / Firefox

These hardware and software requirements ensure smooth development, secure biometric processing, reliable blockchain interaction, and effective system demonstration. The chosen configuration supports a cost-effective yet realistic implementation suitable for academic and institutional election environments.

Chapter-3

SOFTWARE REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) describes the functional and non-functional requirements of the MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System. This document defines what the system will do, who will use it, and the conditions under which it operates. The SRS ensures that the system meets the expectations of all stakeholders, including election administrators, registrars, election officers, and voters, while maintaining security, transparency, and reliability.

3.1 Users

The system supports three authorized user roles, each with defined responsibilities and access privileges.

1. Administrator (Admin)

- Responsible for creating and managing elections.
- Adds and manages candidates for each election.
- Deploys and monitors blockchain smart contracts.
- Views election status, voting progress, and final results.
- Has full control over system configuration and monitoring.

2. Registrar

- Responsible for registering voters in the system.
- Captures voter biometric fingerprints using the fingerprint scanner.
- Ensures that duplicate voter registrations are prevented.
- Stores voter details securely in the local database.
- Does not have access to voting or result management functions.

3. Election Officer

- Verifies voter identity using biometric fingerprint authentication.
- Controls and supervises the voting process during election hours.

- Grants voting access only after successful biometric verification.
- Ensures that each voter is allowed to vote only once.

3.2 Functional Requirements

The functional requirements define the core operations that the MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System must perform to ensure secure, transparent, and reliable digital voting.

1. User Authentication and Authorization

- The system must provide secure login for Administrator, Registrar, and Election Officer.
- Role-based access control must restrict system features according to the logged-in user's role.
- Unauthorized access attempts must be denied by the system.

2. Voter Registration (Registrar Module)

- The system must allow the Registrar to register eligible voters.
- The registrar must be able to capture voter fingerprint data using the biometric scanner.
- The system must verify that a voter is not already registered.
- Only hashed biometric data must be stored in the database.
- The registrar must not have access to voting or result declaration features.

3. Biometric Verification (Election Officer Module)

- The system must allow the Election Officer to verify voter identity using fingerprint authentication.
- The system must compare the live fingerprint input with stored hashed biometric data.
- Voting access must be granted only after successful biometric verification.
- Failed verification attempts must be logged for monitoring purposes.

4. Election Creation and Management (Admin Module)

- The system must allow the Administrator to create elections for different regions or districts.
- The administrator must be able to define election start and end dates.
- The system must automatically update election status (Pending, Active, Completed).

- The administrator must be able to view election details and voting progress.

5. Candidate Management (Admin Module)

- The system must allow the administrator to add, update, and delete candidates.
- Candidates must be linked to a specific election.
- Candidate details must be stored securely in the database.
- Candidate information must be retrievable during the voting process.

6. Voting Process (Officer-Controlled Voting)

- The system must allow a verified voter to cast only one vote.
- Voting must be enabled only by the Election Officer after biometric verification.
- Each vote must be recorded as a blockchain transaction.
- Duplicate voting must be strictly prevented.

7. Blockchain Integration

- The system must deploy and interact with blockchain smart contracts.
- Each vote must be permanently stored as an immutable blockchain transaction.
- The system must support real-time transaction recording and validation.
- Blockchain data must be used as the source of truth for vote counting.

8. Result Computation and Viewing

- The system must automatically calculate election results based on blockchain data.
- The administrator must be able to view final election results.
- Results must reflect accurate vote counts without manual intervention.

9. System Monitoring and Logs

- The system must maintain logs for voter registration, biometric verification, and voting activity.
- The administrator must be able to monitor system activity for audit purposes.

3.3 Non-Functional Requirements

Non-functional requirements ensure quality, performance, and reliability of the system.

1. Performance

- The system must support multiple voters without noticeable delay.
- Biometric verification should complete within a few seconds.
- Blockchain transaction processing must remain consistent during voting.

2. Security

- Biometric data must never be stored in raw format.
- Cryptographic hashing must be used for sensitive data.
- Blockchain immutability must protect votes from tampering.
- Role-based access control must prevent unauthorized actions.

3. Reliability

- The system must function reliably during the entire voting period.
- Vote data must not be lost even in case of system restart.
- Blockchain records must remain consistent and verifiable.

4. Usability

- The user interface must be simple and easy to understand.
- Minimal training should be required for election officials.
- The voting process should be clear and user-friendly for voters.

5. Scalability

- The system design must allow expansion to multiple elections or regions.
- The blockchain architecture must support additional nodes if required.
- The database must support growth in voter and election data.

6. Maintainability

- The system must be modular for easy updates and enhancements.
- Code should be well-structured for debugging and future development.
- System logs must assist in maintenance and troubleshooting.

7. Portability

- The system must run on Windows-based environments.
- The application should be adaptable to other platforms with minimal changes.
- Blockchain components should remain platform-independent.

Chapter-4

SYSTEM DESIGN AND ANALYSIS

4.1 System Architecture

The MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System follows a modular and role-based system architecture designed to ensure security, transparency, and controlled access throughout the election lifecycle. The architecture separates responsibilities among the Admin, Registrar, and Election Officer, while integrating biometric authentication and blockchain-based vote storage.

Architecture Components

1. Admin Module

The Admin module acts as the central authority of the system. It is responsible for:

- Creating login credentials for Registrars and Election Officers.
- Managing elections by defining election details such as name, start date, and end date.
- Adding candidates to the election.
- Initiating result calculation after election completion.

All election and candidate-related information is recorded on the blockchain, ensuring immutability and transparency.

2. Registrar Module

The Registrar module handles voter enrollment before election creation. Its responsibilities include:

- Capturing voter fingerprints using the SecuGen Hamster Pro 20 biometric device.
- Performing fingerprint quality checks and duplicate detection.
- Converting biometric data into secure hash values to protect voter privacy.
- Storing voter registration details securely in the SQLite database.

This ensures that only verified and unique voters exist in the system before any election is conducted.

3. Election Officer Module

The Election Officer module operates during the active election phase. It is responsible for:

- Verifying voter identity using fingerprint authentication.
- Ensuring that each voter votes only once.
- Enabling the voting interface only after successful biometric verification.

Upon vote submission, the vote is transmitted to the blockchain for permanent recording.

4. Biometric Authentication Layer

This layer integrates the fingerprint scanner and biometric processing logic:

- Captures fingerprint data.
 - Performs AI-assisted quality and uniqueness verification.
 - Generates hashed biometric templates instead of storing raw fingerprint images.
- This approach ensures both security and privacy preservation.

5. Blockchain Layer

The blockchain layer is implemented using Ganache (private Ethereum blockchain):

- Smart contracts store election data, candidate details, and votes.
- Each vote is recorded as a blockchain transaction, creating an immutable audit trail.
- Votes cannot be altered or deleted once recorded.

Ganache is used for demonstration and development purposes while maintaining real blockchain architecture and behavior.

6. Database Layer

The SQLite database is used for:

- Storing user credentials and role-based access data.
- Maintaining voter registration and voting status (voted / not voted).
- Supporting fast local access without exposing sensitive biometric data.

7. User Interface Layer

The system provides role-specific dashboards:

- **Admin Dashboard**
- **Registrar Dashboard**
- **Election Officer Dashboard**

Each dashboard exposes only the functions permitted for that role, ensuring controlled access and ease of use.

Architecture Workflow Order

1. Admin creates Registrar and Officer login credentials.
2. Registrar registers voters using biometric authentication.
3. Admin creates elections and adds candidates on the blockchain.
4. Election Officer verifies voters and enables voting.
5. Votes are recorded as blockchain transactions.
6. Admin calculates results directly from blockchain data.

Key Architectural Advantages

- Role-based access control ensures system integrity.
- Biometric hashing preserves voter privacy.
- Blockchain ensures tamper-proof vote storage.
- Modular design allows future enhancements such as public blockchain deployment.

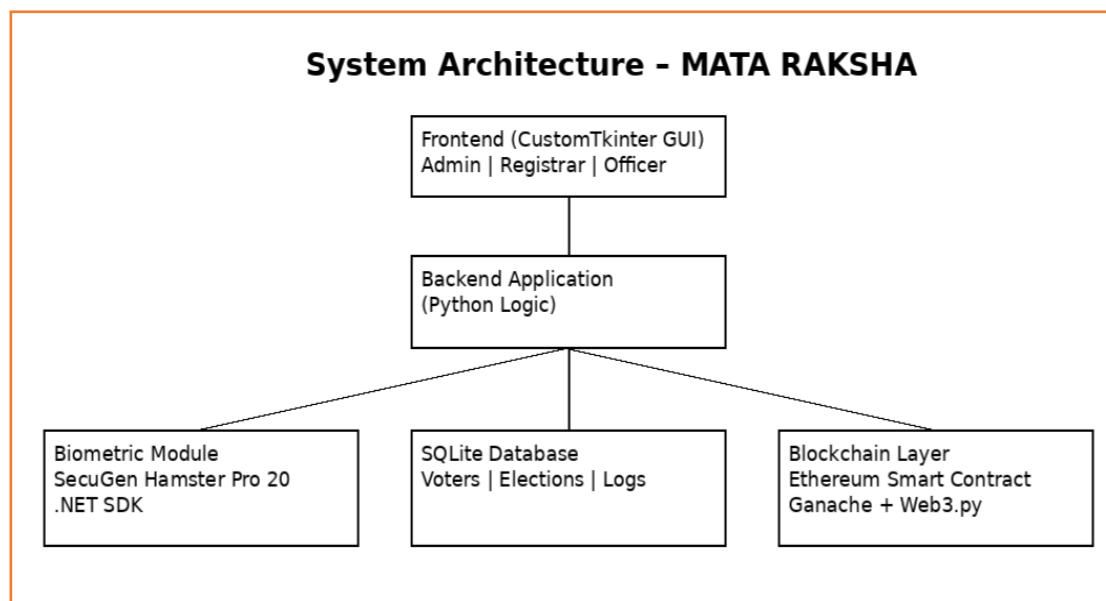


Figure 4.1: System Architecture Diagram

4.2 Flow Diagram

Phase 1: System Initialization (Admin Setup)

The Admin logs into the Admin Dashboard using secure credentials. Admin creates login credentials for the Registrar and Election Officer. Role-based access control ensures that each user can access only their permitted functions.

Outcome:

The system is securely initialized with authorized roles.

Phase 2: Voter Registration (Pre-Election Phase)

The Registrar logs in using the credentials provided by the Admin. Eligible voters are registered by capturing their fingerprints using the SecuGen Hamster Pro 20 device.

The biometric module:

- Performs fingerprint quality validation
- Converts the fingerprint into a secure hash
- Checks for duplicate registrations
- Verified voter details and hashed biometric data are stored in the SQLite database.

Outcome:

Only unique and verified voters exist in the system before election creation.

Phase 3: Election Creation (Post-Registration Phase)

The Admin creates the election by specifying:

- Election name
- Start date
- End date
 - Admin adds candidates for the election.
 - Election and candidate details are registered on the blockchain via smart contracts.

Outcome:

Election setup becomes immutable and transparent.

Phase 4: Voting Phase (Election Officer Controlled)

During the active election window, the Election Officer logs in. A voter arrives at the polling station. The Officer captures the voter's fingerprint for verification.

The system:

- Matches the fingerprint hash with registered data
- Confirms the voter has not already voted
 - Upon successful verification, the voting interface is unlocked.
 - The voter selects their preferred candidate.

Outcome:

Only authenticated voters are allowed to vote once.

Phase 5: Vote Recording (Blockchain Transaction)

The selected vote is transmitted to the blockchain layer (Ganache). A smart contract records the vote as a blockchain transaction. A new block is created containing vote data and timestamp. No voter identity or biometric information is stored on the blockchain.

Outcome:

Votes are stored in an immutable, tamper-proof, and auditable manner.

Phase 6: Election Closure and Result Calculation

After the election end time, voting is automatically disabled. The Admin initiates result calculation. Votes are counted directly from the blockchain, ensuring accuracy. The system

computes:

- Total votes per candidate
- Final winner

Outcome:

Results are transparent and free from manual manipulation.

Phase 7: Audit and Transparency

Blockchain transactions can be viewed using Ganache. Each vote can be verified without revealing voter identity. The system supports post-election auditing and verification.

Outcome:

High trust, transparency, and accountability.

The workflow can be visualized as:

1. Admin Login → Registrar & Officer Credentials Created
2. Registrar Login → Voter Registration with Fingerprint
3. Fingerprint Verified → Voter Data Stored Securely
4. Admin Creates Election → Candidates Added on Blockchain
5. Election Starts → Officer Verifies Voter Biometrically
6. Voter Authenticated → Voting Interface Enabled
7. Vote Cast → Blockchain Transaction Recorded
8. Block Created → Vote Stored Immutably
9. Election Ends → Votes Counted from Blockchain
10. Results Declared → Winner Announced

Complete Workflow of AI-Enhanced Biometric Blockchain Voting System

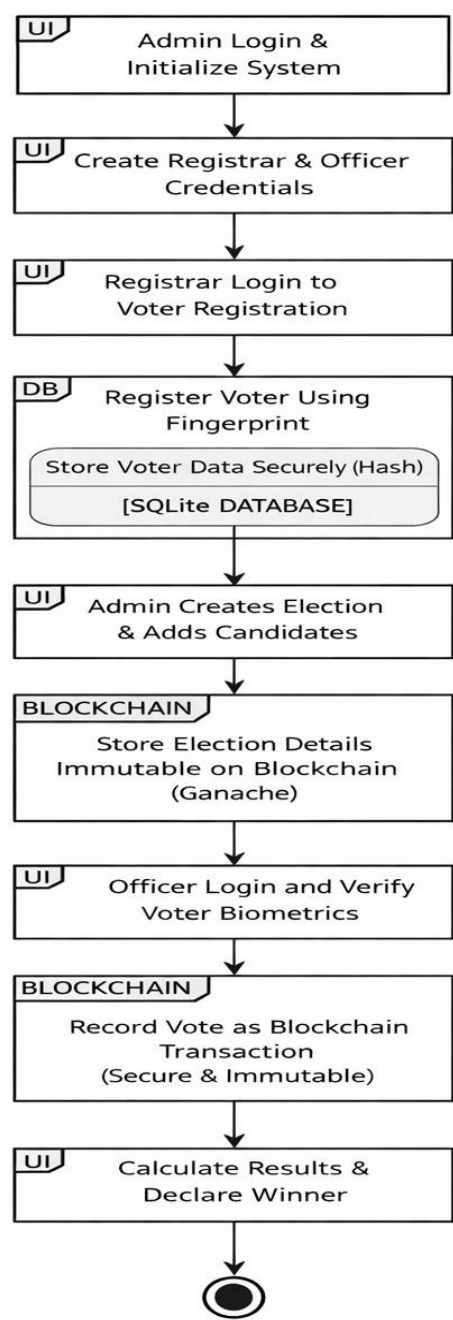


Figure 4.2: Flow diagram

Chapter-5

DETAILED DESIGN

5.1 Use Case Diagram

Actors

1. Administrator (Admin)

- Responsible for overall system control and configuration
- Creates and manages Registrar and Officer accounts
- Creates elections, adds candidates, and controls election lifecycle
- Views complete election results and audit logs

2. Registrar

- Registers voters before the election begins
- Captures voter fingerprint using biometric device
- Stores voter details securely in the database
- Ensures duplicate or invalid voter registration is prevented

3. Election Officer

- Verifies voter identity using fingerprint authentication on polling day
- Enables eligible voters to cast their vote
- Monitors voting progress during the election
- Cannot modify voter or election data

4. Voter (Indirect Actor)

- Does not directly interact with the system UI
- Participates by biometric verification and casting a vote
- Remains anonymous after vote submission

Primary Use Cases (High Level)

Authenticate User

- Admin, Registrar, and Officer log in using secure credentials
- Role-based access ensures users see only permitted dashboards

Manage Users & Roles (Admin)

- Create, update, or disable Registrar and Officer accounts
- Assign role-based permissions

Voter Registration (Registrar)

- Capture voter details and fingerprint
- Validate fingerprint uniqueness
- Store encrypted biometric hash in SQLite database

Election Management (Admin)

- Create elections (district/constituency level)
- Add candidates and configure election parameters
- Activate or close elections

Voter Verification (Officer)

- Authenticate voter fingerprint at polling time
- Confirm voter eligibility and voting status

Cast Vote (Officer-assisted)

- Record vote securely on the blockchain (Ganache)
- Ensure “one voter – one vote” rule
- Maintain voter anonymity

Blockchain Transaction Handling

- Store vote transactions immutably

- Prevent tampering or modification of votes

Result Computation & Declaration (Admin)

- Retrieve vote counts from blockchain
- Generate transparent and verifiable results

Audit & Transparency

- Maintain logs for registration, voting, and results
- Support verification without exposing voter identity

MATA RAKSHA Voting System: Roles and Responsibilities

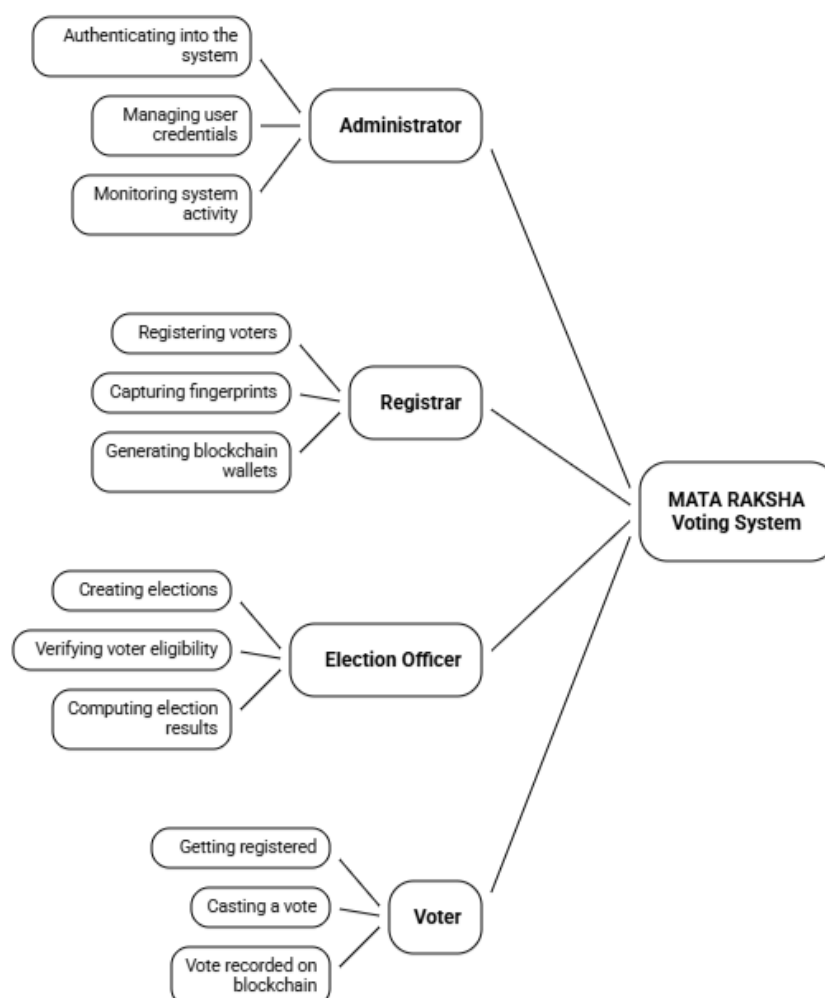


Figure 5.1: Use Case Diagram

5.2 Class Diagram

Design Overview

The class diagram represents the structural design of the MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System. The system follows a normalized user model with clear separation of responsibilities across election management, voter registration, biometric verification, and blockchain-based vote recording.

The design emphasizes security, scalability, and role-based access control, while supporting integration with biometric devices and blockchain infrastructure.

Core Classes and Relationships

1. User

The User class represents all system users in a normalized manner.

Attributes:

- user_id
- username
- password_hash
- role (ADMIN / REGISTRAR / OFFICER)
- district
- created_at

Description:

- Acts as the base class for authentication and authorization.
- Role-based access ensures users can only perform permitted actions.
- Logical normalization improves scalability and simplifies future extensions.

Relationship:

- One User performs multiple system actions depending on role.

2. Voter

Represents a registered voter with biometric and blockchain identity.

Attributes:

- voter_id
- name
- date_of_birth
- aadhaar_number
- district
- fingerprint_hash
- blockchain_address
- registered_at

Description:

- Stores only hashed biometric data to protect privacy.
- Each voter is uniquely mapped to a blockchain wallet.
- Voters are registered before election creation.

Relationship:

- A Registrar (User) registers many Voters (1-to-many).

3. Election

Defines an election conducted in a specific district.

Attributes:

- election_id
- election_name
- district
- start_date
- end_date
- contract_address

- status (Pending / Active / Completed)

Description:

- Each election is deployed as a smart contract on the blockchain.
- Election lifecycle is controlled by authorized users.

Relationship:

- One Election has many Candidates.
- One Election is managed by an Officer (User).

4. Candidate

Represents candidates participating in an election.

Attributes:

- candidate_id
- candidate_name
- party_name
- symbol

Description:

- Candidates are linked to a specific election.
- Candidate data is immutable once the election becomes active.

Relationship:

- Many Candidates belong to one Election (many-to-one).

5. Vote (Blockchain Layer)

Logical representation of vote casting.

Attributes:

- transaction_hash

- election_id
- candidate_id
- timestamp

Description:

- Votes are **not stored in the local database**.
- Each vote is recorded as a blockchain transaction.
- Ensures immutability, transparency, and tamper resistance.

Relationship:

- One Voter casts one vote per Election.
- Votes are validated by smart contracts.

6. Authentication & Security Services

Supporting service classes:

- **AuthService**

Handles login and credential verification.

- **BiometricService**

Captures and validates fingerprint templates.

- **BlockchainService**

Deploys smart contracts and records votes.

- **DatabaseService**

Manages SQLite persistence.

Key Design Highlights

- **Normalized User Model** simplifies authentication and authorization.
- **Biometric data is stored only as hashes**, ensuring privacy.

- **Votes are stored exclusively on blockchain**, eliminating vote manipulation.
- **Clear separation of concerns** between database, biometric processing, and blockchain.
- **Scalable architecture** suitable for district-level and institutional elections.

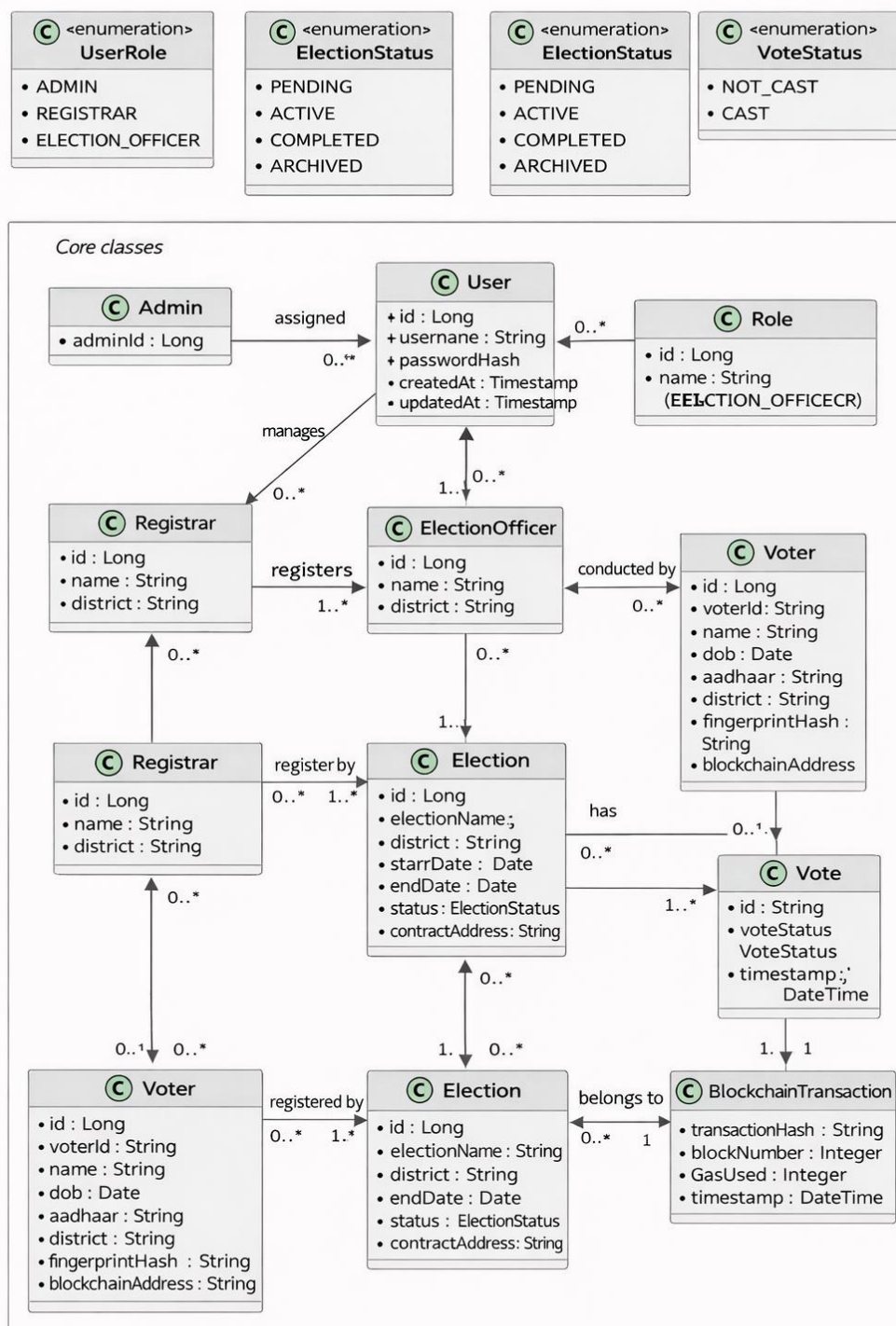


Figure 5.2: Class Diagram

5.3 ER Diagram

Entities & Keys (Relational View)

This section describes the core data entities involved in the *MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System*. The design follows database normalization principles while integrating blockchain-based election integrity. Certain election-specific data (such as vote casting and voter eligibility) are maintained on the blockchain; corresponding relational entities are represented conceptually for clarity and documentation.

1. Users

Purpose:

Stores authentication and role-based access information for system operators.

Entity: Users

Table 5.1 Users Table

Attribute	Description
user_id (PK)	Unique identifier for each user
username	Unique login name
password_hash	Encrypted password
role	Defines user role: ADMIN / REGISTRAR / OFFICER
district	Operational district (role dependent)
created_at	Account creation timestamp

Constraints:

- For **Admin**, district can be NULL.
- For **Registrar**, district must be NOT NULL.
- For **Election Officer**, district must be NOT NULL and UNIQUE.

2. Voters

Purpose:

Maintains verified voter identity information using biometric and demographic data.

Entity: Voters

Table 5.2 Voters Table

Attribute	Description
voter_id (PK)	System-generated voter identifier
name	Voter's full name
dob	Date of birth
aadhaar	Unique national ID number
district	Voter's district
fingerprint_hash	Secure biometric hash
blockchain_address	Unique blockchain wallet address
registered_at	Registration timestamp

3. Elections

Purpose:

Stores election metadata and links off-chain records with blockchain smart contracts.

Entity: Elections

Table 5.3 Elections Table

Attribute	Description
election_id (PK)	Unique election identifier
election_name	Name of the election
district	Election district
start_date	Election start date
end_date	Election end date
contract_address	Deployed blockchain contract address
status	Pending / Active / Completed
created_at	Creation timestamp

4. Candidates

Purpose:

Stores candidate details for each election.

Entity: Candidates

Table 5.4 Candidates Table

Attribute	Description
candidate_id (PK)	Unique candidate identifier
election_id (FK)	References Elections
candidate_name	Candidate name
party_name	Political party
symbol	Party symbol

5. Vote Records (Conceptual – Blockchain Controlled)

Purpose:

Vote data is **not stored in the database** to prevent tampering. All vote casting and counting are handled exclusively by blockchain smart contracts.

Blockchain Representation:

- hasVoted[address]
- VoteCast event
- Immutable transaction records

6. Election–Voter Registration (Conceptual)

Purpose:

Represents voter eligibility for a specific election, enforced by the blockchain.

Entity: Election_Voter_Registration (*Conceptual*)

Table 5.5 Election Voter Registration BC

Attribute	Description
election_id (FK)	Election reference
voter_blockchain_address	Voter wallet address
registered_by	Admin who registered voter
registered_at	Timestamp

Primary Key:

(election_id, voter_blockchain_address)

Note:

This maps directly to the smart contract's voter registration logic.

7. Election–Vote Status (Conceptual)**Purpose:**

Tracks whether a voter has cast a vote in a particular election.

Entity: Election_Vote_Status (*Conceptual*)

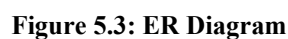
Table 5.6 Election Vote Status BC

Attribute	Description
election_id (FK)	Election reference
voter_blockchain_address	Voter wallet address
has_voted	Boolean vote flag
voted_at	Block timestamp
transaction_hash	Blockchain transaction ID

Primary Key:

(election_id, voter_blockchain_address)

- The database is normalized up to Third Normal Form (3NF).
- Vote casting and result integrity are ensured through blockchain immutability.
- Conceptual relational entities are used to explain blockchain mappings without duplicating sensitive vote data in the database.
- This hybrid architecture ensures security, transparency, scalability, and auditability.



Chapter-6

IMPLEMENTATION

6.1 Modules

The MATA RAKSHA system is divided into modular components to ensure security, transparency, and intelligence in the election process. Artificial Intelligence (AI) is used to enhance biometric verification, detect anomalies, and improve decision reliability.

1. Authentication and Authorization Module

- Handles secure login for Admin, Registrar, and Election Officer.
- Uses encrypted password verification and role-based access control.
- Prevents unauthorized access to election and voter data.
- Ensures strict separation of responsibilities among system users.

Outcome:

Only authorized officials can access system functionalities.

2. User & Role Management Module (Admin)

- Admin creates and manages login credentials for Registrars and Officers.
- Assigns role-based and district-based access.
- Enforces access rules such as:
 - Admin → system-wide access
 - Registrar → voter registration only
 - Officer → election and result management

Outcome:

Controlled delegation and secure governance.

3. Voter Registration Module (Registrar)

- Registrar registers voters before election creation.
- Captures voter demographic details and biometric fingerprint data.
- Fingerprint is converted into a secure template and hash.

- A blockchain wallet address is generated for each voter.

Outcome:

Only eligible and verified voters are enrolled.

4. AI-Based Biometric Verification Module

- Uses AI-assisted fingerprint quality analysis before verification.
- AI checks:
 - Image clarity
 - Noise level
 - Fingerprint completeness
- Rejects poor-quality biometric samples to prevent false matches.
- Matches fingerprint hash with registered biometric template.

AI Role:

Improves accuracy, reduces false acceptance/rejection, and enhances system reliability.

Outcome:

Accurate and intelligent voter authentication.

5. Election Management Module (Election Officer)

- Officer creates elections for assigned districts.
- Manages election lifecycle:
 - Pending → Active → Completed → Archived
- Adds and manages candidates.
- Controls voting window strictly by date and time.

Outcome:

Structured and controlled election execution.

6. Blockchain Voting Module

- Votes are recorded as blockchain transactions using smart contracts.
- Ensures:

- One person → One vote
 - Immutability of vote data
- Uses Ganache (private blockchain) for development and demonstration.
- Blockchain logic remains identical to real Ethereum deployment.

Outcome:

Tamper-proof and transparent voting mechanism.

7. AI-Based Anomaly Detection Module

- AI logic monitors voting behavior patterns such as:
 - Multiple failed biometric attempts
 - Rapid repeated access attempts
- Flags suspicious activities for review.
- Helps identify potential misuse or system abuse.

AI Role:

Enhances election security by early detection of abnormal behavior.

Outcome:

Proactive fraud prevention.

8. Result Computation & Declaration Module

- Fetches vote counts directly from the blockchain.
- Computes results automatically after election completion.
- Eliminates manual counting and human bias.
- Displays final results securely.

Outcome:

Accurate, fast, and trustworthy result declaration.

9. Audit & Transparency Module

- Provides immutable blockchain transaction hashes.
- Enables verification without revealing voter identity.

- Supports post-election audits and integrity checks.

Outcome:

Public trust through verifiable transparency.

6.2 Algorithm

Algorithm 1: User Authentication Algorithm

Input: Username, Password

Output: Authenticated session with role access

Steps:

1. User enters login credentials.
2. System verifies password using secure hashing.
3. If valid, user role is identified (Admin / Registrar / Officer).
4. Access to corresponding dashboard is granted.
5. Invalid credentials result in access denial.

Algorithm 2: AI-Assisted Voter Registration Algorithm

Input: Voter details and fingerprint image

Output: Registered voter profile

Steps:

1. Registrar enters voter details.
2. Fingerprint is captured.
3. AI checks fingerprint quality.
4. Valid fingerprint is converted into a secure hash.
5. Blockchain wallet is generated.
6. Voter record is stored successfully.

Algorithm 3: Vote Casting Algorithm (Blockchain-Based)

Input: Verified voter, selected candidate

Output: Blockchain vote transaction

Steps:

1. Voter undergoes biometric verification.
2. System checks voting eligibility.
3. Voter selects a candidate.
4. Vote is recorded as a blockchain transaction.
5. Transaction hash is generated and stored.
6. System marks voter as “voted” to prevent duplicate voting.

Algorithm 4: Result Computation Algorithm

Input: Blockchain vote records

Output: Election result

Steps:

1. Election is marked as completed by Officer.
2. Smart contract tallies votes for each candidate.
3. Results are fetched from blockchain.
4. Winner and vote counts are displayed.
5. Results are locked to prevent changes.

Algorithm 5: Audit Verification Algorithm

Input: Election ID

Output: Verified audit trail

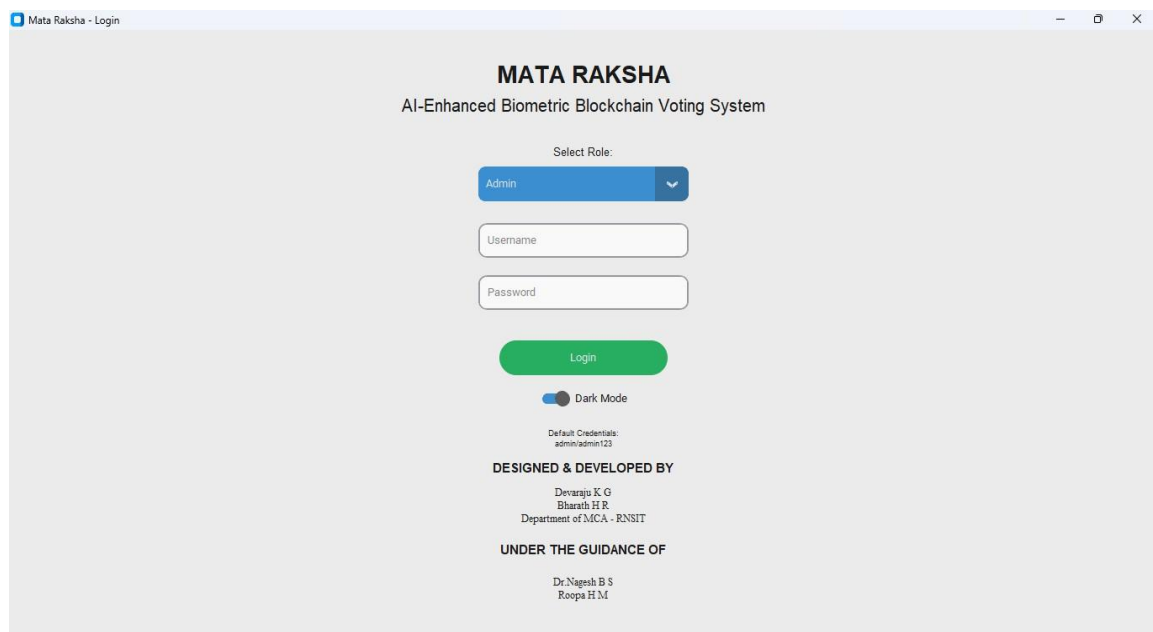
Steps:

1. System retrieves blockchain transactions for the election.

2. Validates transaction integrity.
3. Displays transaction hashes for verification.
4. Ensures vote counts match blockchain records.

6.3 Screen-shots

1. Login window: This screen allows users to log in by selecting their role (Admin, Registrar, or Officer) and entering valid credentials. It ensures secure, role-based access to the system.



The screenshot shows the login interface for the MATA RAKSHA system. The window title is "Mata Raksha - Login". The main heading is "MATA RAKSHA" with the subtitle "AI-Enhanced Biometric Blockchain Voting System". Below this, there is a "Select Role:" dropdown menu currently set to "Admin". Underneath the dropdown are two input fields labeled "Username" and "Password". A green "Login" button is positioned below the password field. A "Dark Mode" toggle switch is located below the login button. At the bottom, the default credentials are listed as "admin/admin123". The footer credits the designers and developers as "Devaraju K.G" and "Bharath H.R." from the "Department of MCA - RNSIT", and acknowledges guidance from "Dr. Nagesh B.S" and "Roopa H.M."

Figure 6.1 Login Window

2. Admin Dashboard

Registrar Tab: This module enables the Admin to add, update, and delete Registrar accounts district-wise. It ensures that voter registration is handled only by authorized registrars for each district.

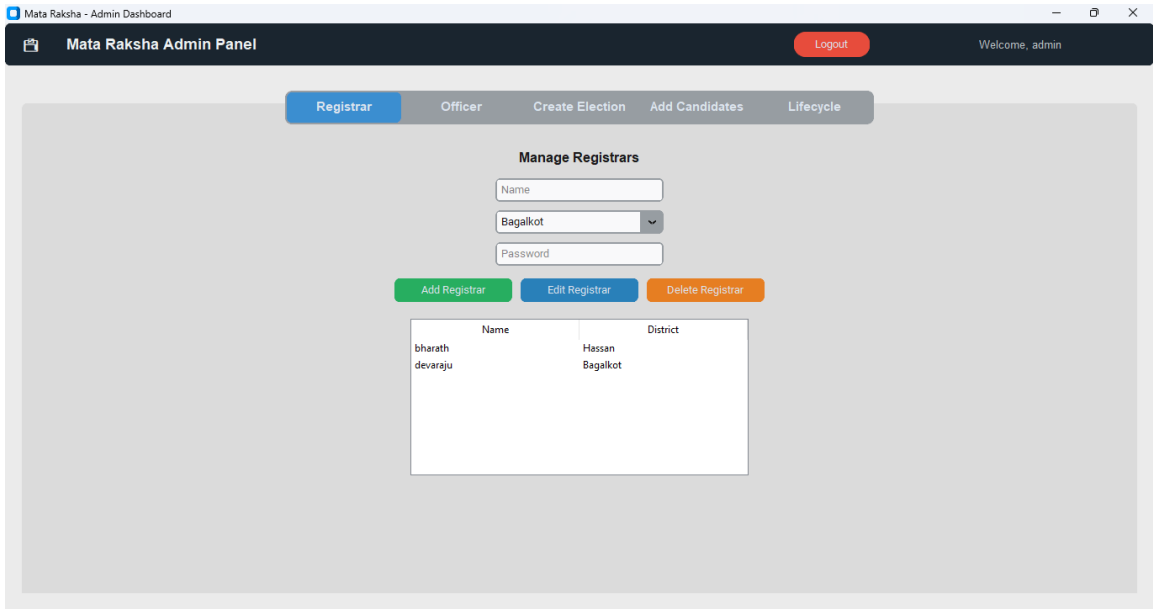


Figure 6.2 Registrar Tab

Officer Tab: This screen allows the Admin to manage Election Officer accounts with district-level assignment. Each officer is uniquely mapped to a district to ensure controlled and secure voting operations.

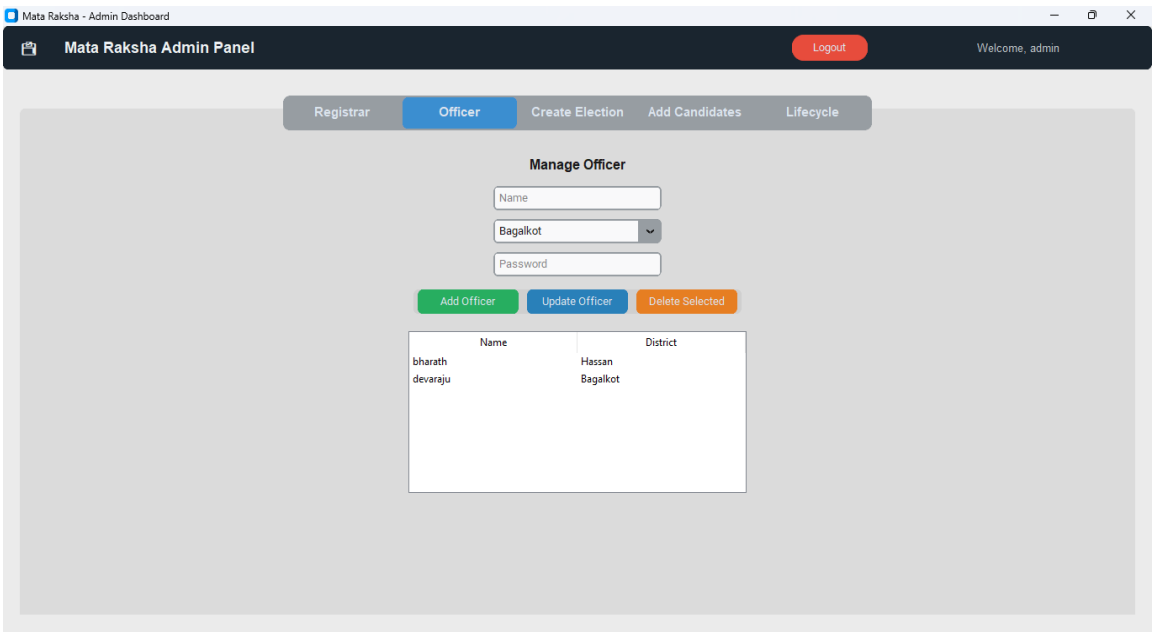


Figure 6.3 Officer Tab

Create Election Tab: The Admin creates elections by defining the election name, start date, and end date. The system supports creating elections across all Karnataka districts with centralized control.

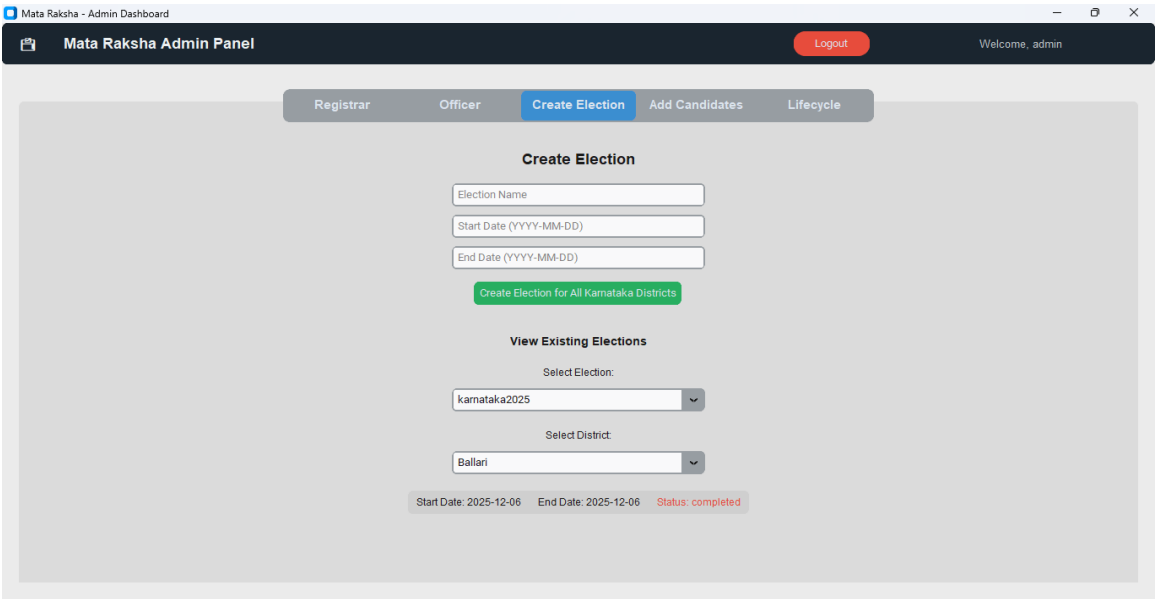


Figure 6.4 Create Election Tab

Add Candidates Tab: This module allows the Admin to add, update, or remove candidates for a selected election and district. Candidate details such as name, party, and election symbol are securely stored and displayed.

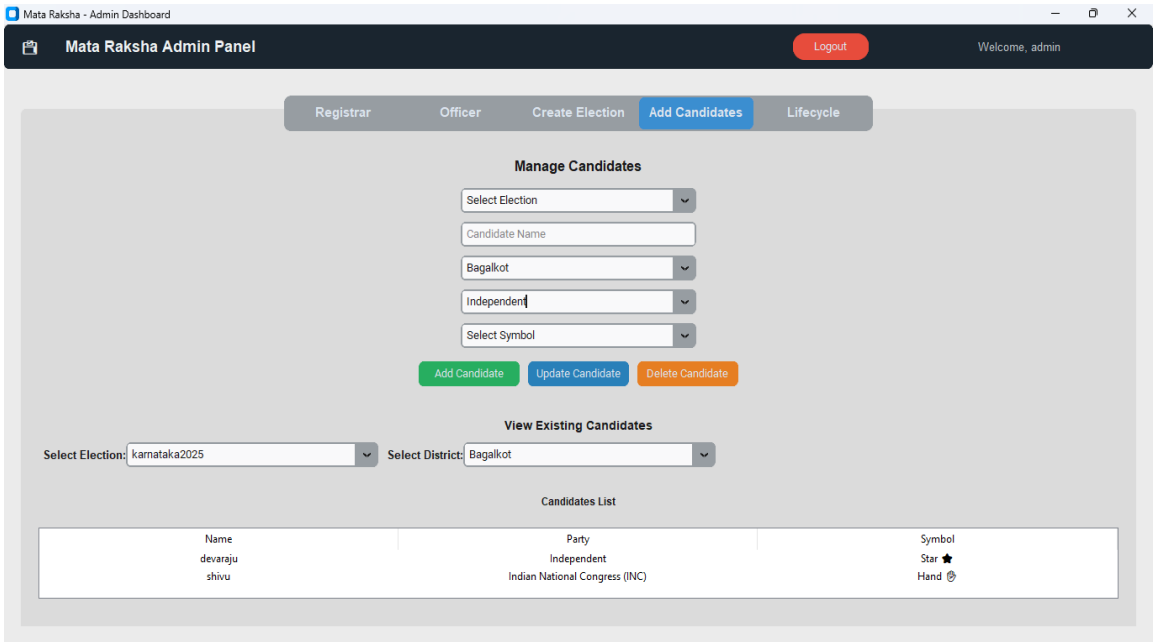


Figure 6.5 Add Candidates Tab

Lifecycle Tab: This screen shows the live election status, voting progress, and blockchain connectivity. After voting ends, the Admin can finalize the election and view transparent, tamper-proof results.

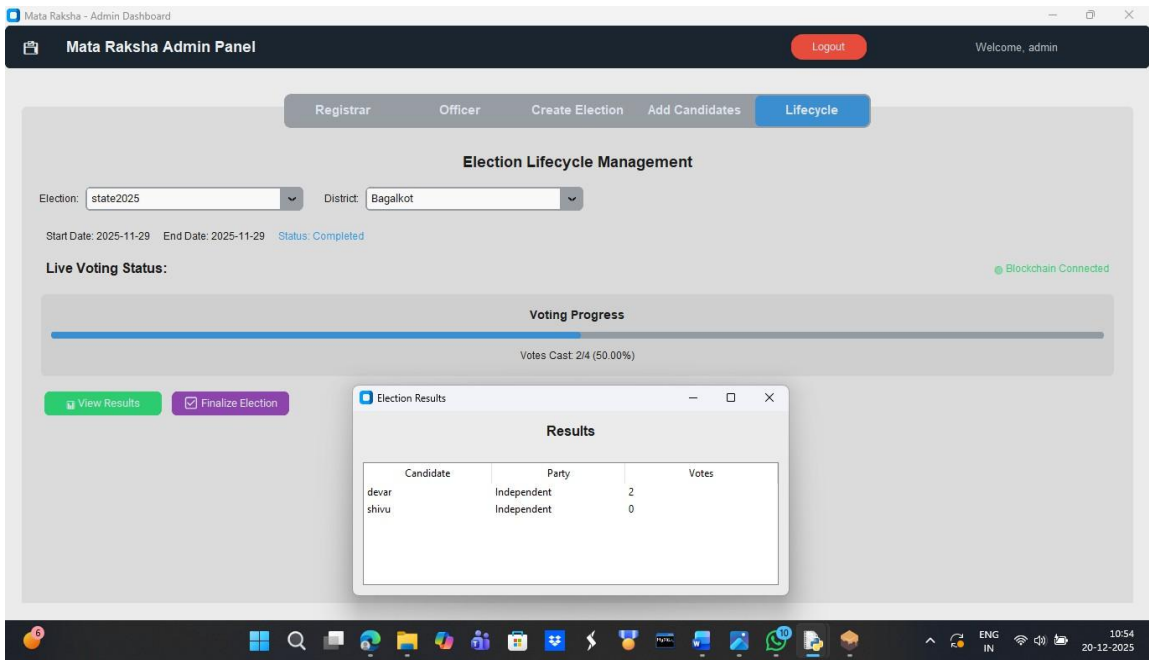


Figure 6.6 Lifecycle Tab

3. Registrar Dashboard

Register Voter Tab: The Registrar registers voters by entering demographic details and capturing fingerprints using a biometric scanner. Each voter is assigned a unique Voter ID, and biometric data is securely stored as a hash.

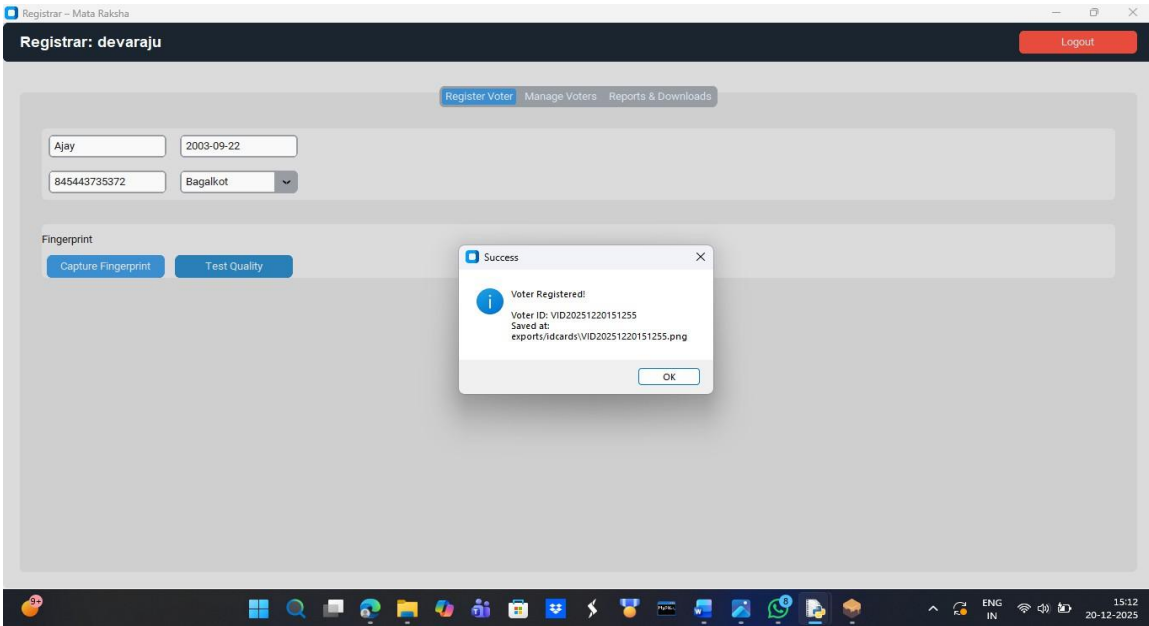
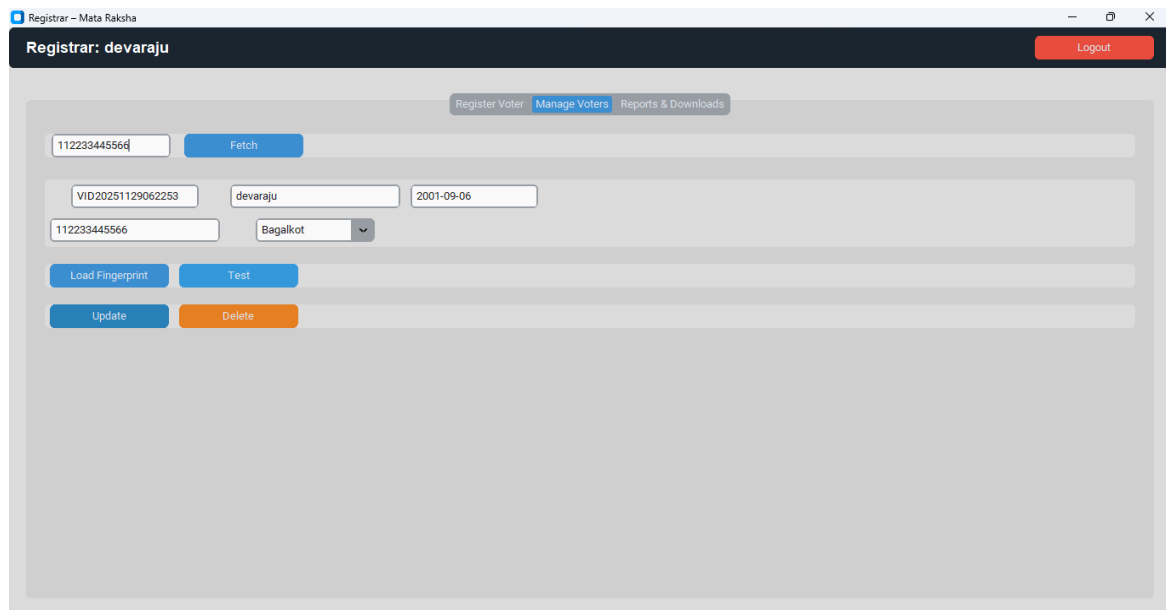


Figure 6.7 Register Voter Tab

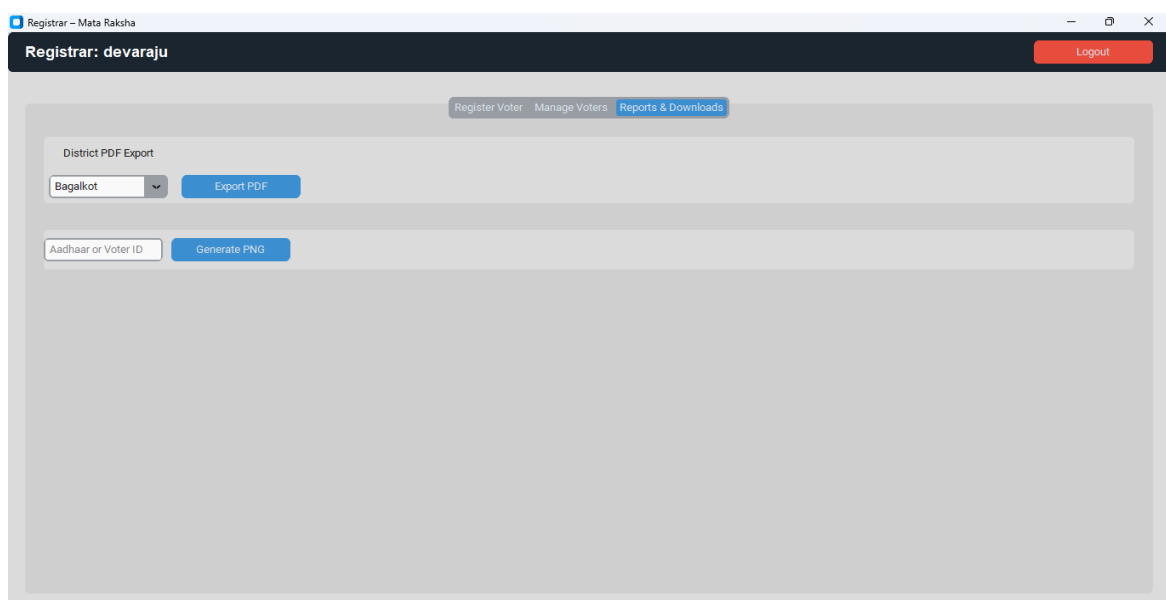
Manage Voters Tab: This screen allows the Registrar to fetch, update, or delete existing voter records. Fingerprint verification ensures that voter data updates are accurate and duplication is prevented.



The screenshot shows the 'Manage Voters' tab in the Registrar's interface. The header bar includes the title 'Registrar: devaraju' and a 'Logout' button. Below the header, there are three tabs: 'Register Voter', 'Manage Voters' (which is active), and 'Reports & Downloads'. The main content area contains several input fields and buttons. At the top, there is a text input field with the value '112233445566' and a 'Fetch' button. Below this, there are three input fields: 'VID20251129062253', 'devaraju', and '2001-09-06'. Further down, there is another text input field with '112233445566' and a dropdown menu showing 'Bagalkot'. At the bottom of the form, there are four buttons: 'Load Fingerprint', 'Test', 'Update', and 'Delete'.

Figure 6.8 Manage Voters Tab

Reports & Downloads Tab: The Registrar can export district-wise voter lists as PDFs and generate voter ID cards as PNG files. This module supports administrative reporting and official documentation needs.



The screenshot shows the 'Reports & Downloads' tab in the Registrar's interface. The header bar is the same as in Figure 6.8, with the title 'Registrar: devaraju' and a 'Logout' button. The tabs are 'Register Voter', 'Manage Voters', and 'Reports & Downloads' (which is active). The main content area features two sections. The first section, titled 'District PDF Export', has a dropdown menu showing 'Bagalkot' and an 'Export PDF' button. The second section has a text input field with 'Aadhaar or Voter ID' and a 'Generate PNG' button.

Figure 6.9 Reports & Downloads Tab

4. Officer Dashboard

Voting & Verification Tab: The Election Officer verifies voter identity using Aadhaar/Voter ID and biometric authentication. Once verified, a secure voting window opens for the voter to cast their vote confidentially.

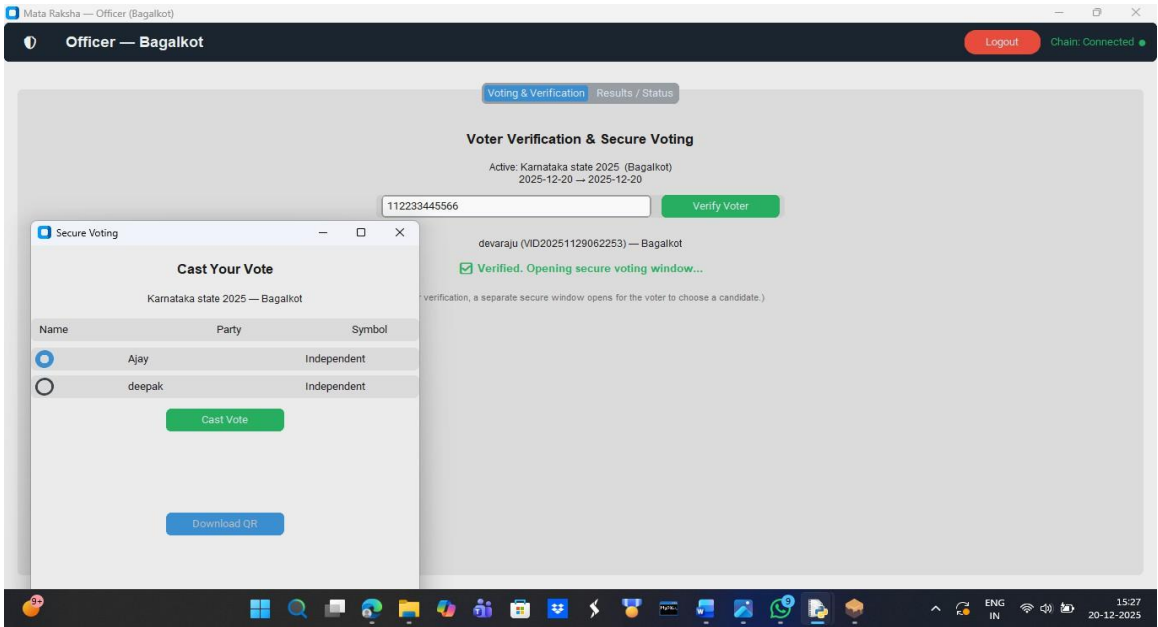


Figure 6.10 Voting & Verification Tab

Results / Status Tab: This screen displays the real-time election status for the selected district, including voter turnout and blockchain connectivity. Election Officers can monitor live voting progress securely and refresh results without accessing individual voter data.

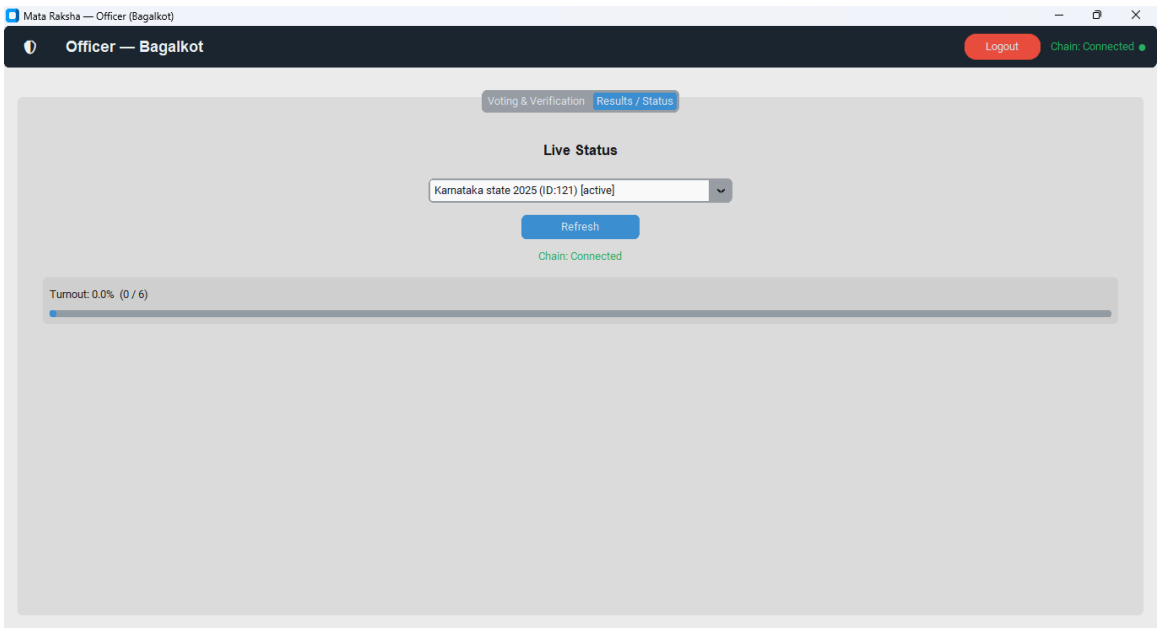


Figure 6.11 Results / Status Tab

Chapter-7

SOFTWARE TESTING

7.1 Types of Tests conducted

To ensure the reliability, security, and correctness of the MATA RAKSHA voting system, multiple levels of testing were carried out. Each test focuses on validating different aspects of the system.

1. Unit Testing

Goal:

To test individual modules and functions independently to ensure they work correctly in isolation.

Examples in our project:

- Verifying user login validation logic.
- Testing voter registration functions to ensure voter details are stored correctly.
- Checking fingerprint hash generation and storage logic.
- Validating smart contract functions for vote casting in Ganache.

2. Integration Testing

Goal:

To verify that different modules interact correctly with each other.

Examples in our project:

- Ensuring voter biometric verification works correctly with the blockchain voting module.
- Verifying that voter registration data flows correctly from the Registrar dashboard to the database.
- Testing interaction between SQLite database and blockchain components during vote casting.
- Ensuring election status changes reflect correctly across Officer and Admin dashboards.

3. Functional Testing

Goal:

To confirm that the system meets all specified functional requirements.

Examples in our project:

- Validating role-based access (Admin, Registrar, Election Officer).
- Ensuring only registered voters can cast votes.
- Verifying that each voter can vote only once.
- Checking correct vote count computation after election completion.
- Ensuring election start and end dates are strictly enforced.

4. System Testing**Goal:**

To test the complete system as a whole in a real-world operating environment.

Examples in our project:

- Running the full application on a Windows system with Python, SQLite, SecuGen SDK, and Ganache.
- Verifying smooth operation of biometric capture, AI verification, blockchain transactions, and dashboards together.
- Ensuring stable performance during multiple voter registrations and vote casting.

5. Validation Testing**Goal:**

To confirm that the system satisfies user and stakeholder requirements.

Examples in our project:

- Ensuring Registrars can register voters easily before election creation.
- Confirming Election Officers can manage elections and declare results without technical complexity.
- Verifying that voters experience a simple and secure voting process.
- Ensuring transparency and auditability meet institutional and electoral expectations.

6. AI Model Validation Testing

Goal:

To validate the effectiveness of AI-assisted biometric quality analysis.

Examples in our project:

- Testing fingerprint quality detection for clarity and noise.
- Ensuring low-quality fingerprints are rejected before verification.
- Verifying AI reduces false acceptance and rejection rates during authentication.

7. Blockchain Testing

Goal:

To ensure correctness and integrity of blockchain-based vote recording.

Examples in our project:

- Verifying vote transactions are immutably recorded in Ganache.
- Ensuring smart contracts prevent duplicate voting.
- Checking accurate vote tally retrieval from blockchain data.

8. User Interface (UI) Testing

Goal:

To evaluate usability, clarity, and responsiveness of dashboards.

Examples in our project:

- Testing Admin, Registrar, and Officer dashboards for correct navigation.
- Ensuring forms (login, voter registration, election creation) work correctly.
- Checking error messages for invalid actions (duplicate voting, invalid dates).
- Ensuring UI remains clear and usable without training.

9. Security Testing

Goal:

To ensure protection against unauthorized access and misuse.

Examples in our project:

- Testing unauthorized role access attempts.
- Verifying secure password storage.
- Ensuring blockchain prevents vote manipulation.
- Validating that voter identity is not exposed during voting.

7.2 Test Cases

The following test cases were executed to validate the system's functionality:

Table 7.1 Test Cases

Test Case ID	Module	Test Description	Input	Expected Output	Result
TC01	Authentication	Test user login with valid credentials	Username: admin, Password: valid	Login successful, redirected to Admin dashboard	Pass
TC02	Authentication	Test login with invalid credentials	Username: wrong, Password: invalid	Error message: "Invalid username or password"	Pass
TC03	User Management	Create Registrar login by Admin	Name: Registrar1, District: Mysuru	Registrar account created successfully	Pass
TC04	User Management	Create Election Officer login	Name: Officer1, District: Mandya	Officer account created, district assigned uniquely	Pass
TC05	Voter Registration	Register voter with valid details	Voter details + fingerprint	Voter registered successfully with blockchain address	Pass
TC06	Voter Registration	Prevent duplicate voter registration	Same Aadhaar / fingerprint	Error shown: "Voter already exists"	Pass

Test Case ID	Module	Test Description	Input	Expected Output	Result
TC07	Election Management	Create election with valid dates	Election name, district, start & end date	Election created and linked to blockchain contract	Pass
TC08	Candidate Management	Add candidate to election	Candidate name, party, symbol	Candidate added and visible in candidate list	Pass
TC09	Biometric Verification	Verify voter using fingerprint	Live fingerprint scan	Fingerprint matched, voting window enabled	Pass
TC10	Voting	Cast vote successfully	Verified voter selects candidate	Vote recorded on blockchain, transaction hash generated	Pass
TC11	Voting	Prevent double voting	Same voter tries to vote again	Voting denied with message “Already voted”	Pass
TC12	Blockchain	Validate blockchain transaction	Election ID	Vote stored immutably in Ganache blockchain	Pass
TC13	Result Management	Compute election results	Completed election	Correct vote count and winner displayed	Pass
TC14	Election Control	Enforce election time window	Vote attempt after end date	Voting blocked, status = Completed	Pass
TC15	Security	Unauthorized access attempt	Registrar tries Admin action	Access denied	Pass

7.3 Test Results and Analysis

Testing was conducted across all major modules of the MATA RAKSHA – AI-Enhanced Biometric Blockchain Voting System, including Authentication, User Management, Voter Registration,

Biometric Verification, Election Management, Blockchain Voting, Result Computation, and Audit Modules. The test outcomes confirm that the system meets both the functional and non-functional requirements defined in the Software Requirements Specification (SRS).

Summary of Results

• Authentication & Authorization:

All test cases related to user login and role-based access control passed successfully. Valid credentials redirected users to the appropriate dashboards, while invalid login attempts displayed correct error messages. This ensured secure and restricted access for Admin, Registrar, and Election Officer roles.

• User & Role Management:

Admin-level operations such as creating Registrar and Election Officer accounts functioned correctly. District-based constraints were enforced as expected, ensuring that Registrars and Officers were assigned to appropriate districts without conflicts.

• Voter Registration Module:

Voter registration tests confirmed that voter details, biometric fingerprint hashes, and blockchain wallet addresses were stored correctly. Duplicate registration attempts using the same Aadhaar number or fingerprint were successfully detected and blocked.

• AI-Based Biometric Verification:

AI-assisted fingerprint quality checks worked as intended by rejecting low-quality or unclear fingerprint inputs. Verified fingerprints were matched accurately, improving authentication reliability and reducing false acceptance or rejection.

• Election Management:

Election creation, candidate addition, and election lifecycle management (Pending → Active → Completed → Archived) operated correctly. Voting was allowed strictly within the defined election time window.

• Blockchain Voting Module:

Votes were successfully recorded as blockchain transactions in Ganache. Each vote generated a unique transaction hash, and the system prevented double voting by the same voter. Blockchain immutability ensured that votes could not be altered or deleted.

- **Result Computation:**

Vote counting and result generation were accurately performed using blockchain data. Final results were displayed correctly after election completion, ensuring transparency and eliminating manual counting errors.

- **Audit & Transparency:**

Blockchain transaction hashes were available for verification, enabling post-election auditing without revealing voter identity. The audit process confirmed consistency between vote counts and blockchain records.

- **User Interface & Usability:**

The Admin, Registrar, and Election Officer dashboards were tested for usability and clarity. Forms, validations, and navigation worked smoothly, providing a user-friendly experience without requiring extensive training.

Overall Analysis

The testing results indicate that the MATA RAKSHA system is secure, reliable, and functionally correct. The integration of AI for biometric verification and blockchain for vote immutability significantly strengthens system trustworthiness. All modules performed as expected, demonstrating that the system is suitable for real-world election scenarios and institutional adoption.

Chapter-8

CONCLUSION

The MATA RAKSHA – AI Enhanced Biometric Blockchain Voting System successfully demonstrates a secure, transparent, and intelligent approach to digital voting. By integrating biometric authentication, artificial intelligence, and blockchain technology, the system effectively addresses key challenges of traditional voting systems such as identity fraud, vote tampering, and lack of transparency.

The use of AI-assisted biometric verification improves the accuracy and reliability of voter authentication while preserving privacy through fingerprint hashing. Blockchain technology ensures that votes are recorded as immutable transactions, making the system tamper-proof and auditable. The role-based design involving Admin, Registrar, and Election Officer ensures proper governance and controlled access at every stage of the election process.

Comprehensive testing across all modules confirms that the system meets both functional and non-functional requirements. The modular architecture, combined with a hybrid database–blockchain approach, makes the system scalable, secure, and suitable for real-world adoption in institutional or governmental election scenarios.

Overall, the project proves that combining AI and blockchain can significantly enhance trust, security, and efficiency in electronic voting systems, making MATA RAKSHA a strong foundation for future digital democracy solutions.

Chapter-9

FUTURE ENHANCEMENTS

Although the proposed system provides a secure and transparent voting mechanism using biometric authentication and blockchain technology, several advanced features can be incorporated in the future to further enhance its robustness, scalability, and real-world applicability.

1. Integration of Zero Knowledge Proofs (ZKP)

In future versions, Zero Knowledge Proofs can be integrated to verify voter eligibility without revealing voter identity or biometric information. This will significantly enhance voter privacy while maintaining authentication accuracy, making the system compliant with advanced privacy standards.

2. Post-Quantum Cryptographic Security

To protect the system against future quantum computing threats, post-quantum cryptographic algorithms can be adopted. This will ensure long-term security of stored votes and blockchain transactions.

3. Real Blockchain Deployment (Public/Test Network)

The current system uses Ganache for development and demonstration. In the future, the system can be deployed on a real blockchain network such as Ethereum testnet or a government-approved private blockchain to support large-scale elections.

4. Offline Voting with Secure Synchronization

An offline voting mechanism can be introduced where votes are temporarily stored in encrypted form and securely synchronized with the blockchain once network connectivity is restored. This enhancement will make the system suitable for remote and rural regions.

5. AI-Based Fraud and Anomaly Detection

Artificial Intelligence models can be added to analyze voting patterns and detect anomalies such as duplicate attempts, abnormal voting behavior, or system misuse, providing an additional layer of intelligent security.

6. Mobile and Web-Based Voting Interfaces

The system can be extended to support secure mobile and web voting platforms using biometric authentication, enabling wider accessibility while maintaining security.

7. Multi-Factor Authentication (MFA)

Additional authentication factors such as OTP or smart card verification can be combined with biometric verification to further strengthen voter authentication.

8. Government-Scale Deployment and Integration

In future, the system can be integrated with national identity platforms and government databases under strict legal frameworks to support large-scale institutional or governmental elections.

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